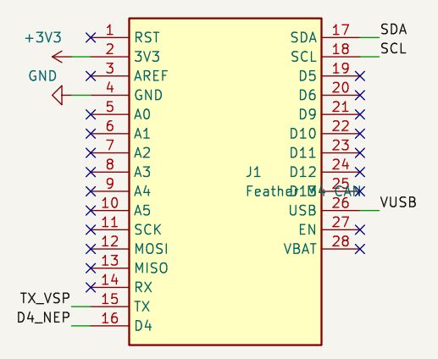
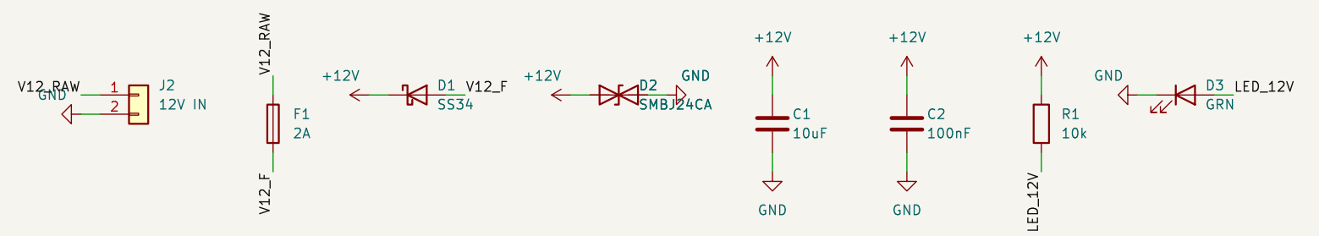
Bench stimulus FeatherWing: 12 V power, SAS power + CAN pass-through, EPS NEP/VSP/torque stimulus

Every EPS line goes through a relay whose NC contact is the car's own wire: no power, no ARM loop or no enable bit means the EPS sees the car exactly as without this board. ARM loop on J5 (pins 7–8): wire an NC e-stop or key switch. Faking VT3/VT6 makes the motor steer: wheels off the ground, hands clear. All enables come from U4 (TCA9534), which powers up with every pin an input; 100k pull-downs hold every driver off until firmware sets the bit.

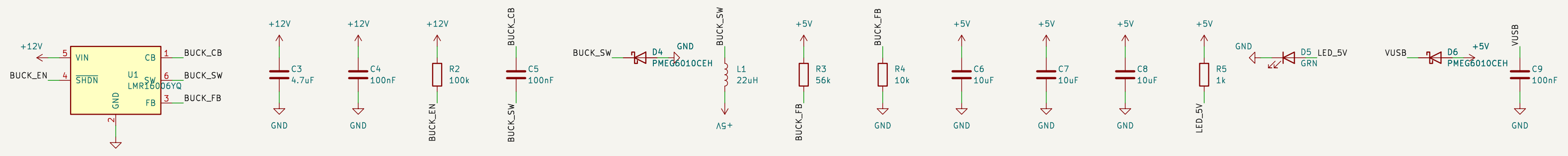
Feather M4 CAN (stacking headers)



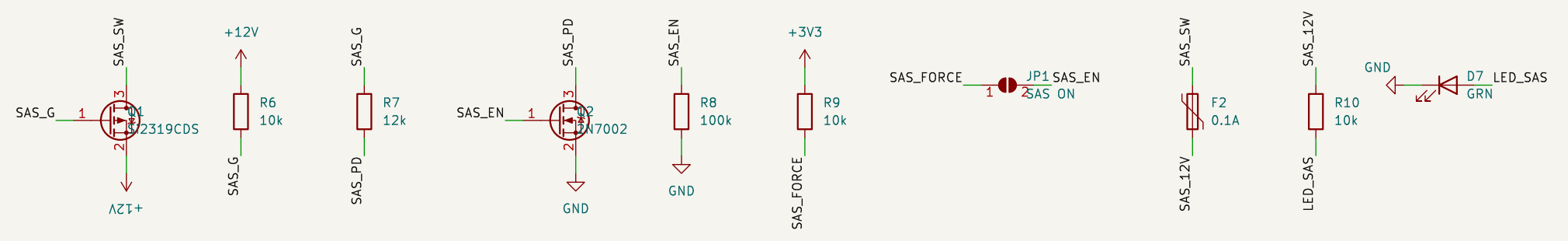
12 V input: fuse, reverse–polarity diode, load–dump TVS


V12_RAW
GND
J2
12V IN
V12_RAW
F1
2A
V12_F
+12V
D1
V12_F
SS34
+12V
D2
SMBJ24CA
GND
+12V
C1
10uF
GND
+12V
C2
100nF
GND
+12V
R1
10k
LED_12V
GND

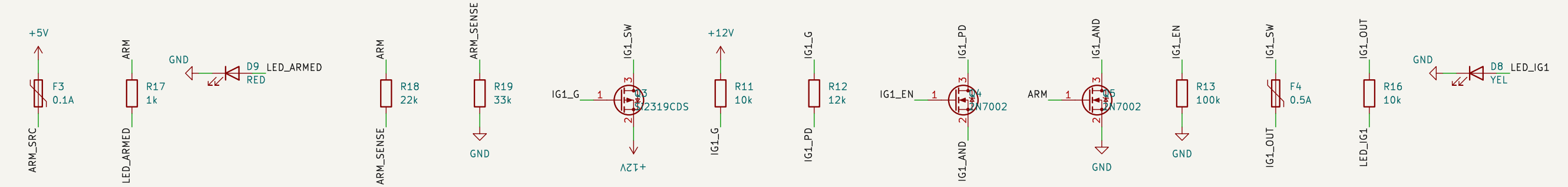
5 V buck (LMR16006, 60 V) and 5 V into the Feather USB pin


+12V
VIN
BUCK_EN
SHDN
U1
LMR16006YQ
GND
CB
SW
FB
BUCK_CB
BUCK_SW
BUCK_FB
GND
+12V
C3
4.7uF
GND
+12V
C4
100nF
GND
+12V
R2
100k
BUCK_EN
BUCK_CB
C5
100nF
BUCK_SW
BUCK_SW
D4
PMEG6010CEH
GND

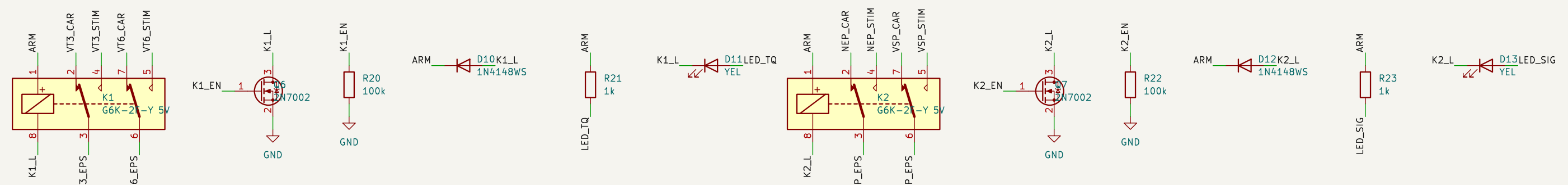
SAS power (12 V, default off; JP1 = always on)


SAS_SW
SAS_G
Q1
SI2319CDS
+12V
+12V
R6
10k
SAS_G
SAS_G
R7
12k
SAS_PD
SAS_PD
SAS_EN
Q2
2N7002
GND
SAS_EN
R8
100k
GND
+3V3
R9
10k
SAS_FORCE
SAS_FORCE
JP1
SAS_EN
SAS ON

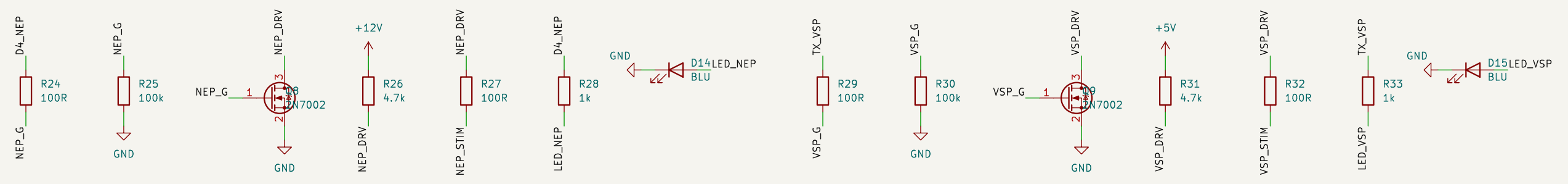
ARM loop (hard kill) and IG1 output (needs IG1_EN and ARM)


+5V
F3
0.1A
ARM_SRC
ARM
R17
1k
LED_ARMED
GND
D9
LED_ARMED
RED
ARM
R18
22k
ARM_SENSE
ARM_SENSE
R19
33k
GND
IG1_SW
IG1_G
Q3
SI2319CDS
+12V
+12V
R11
10k
IG1_G
IG1_G
R12
12k
IG1_PD

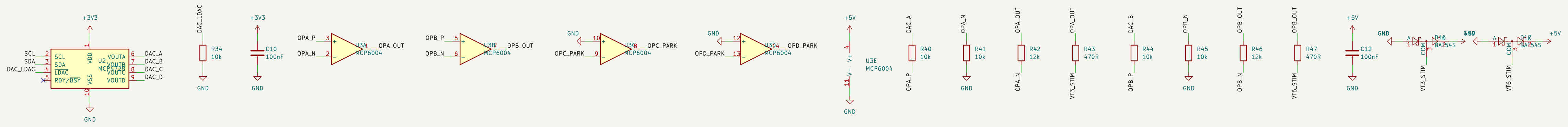
Relays: de–energised = car wiring connected



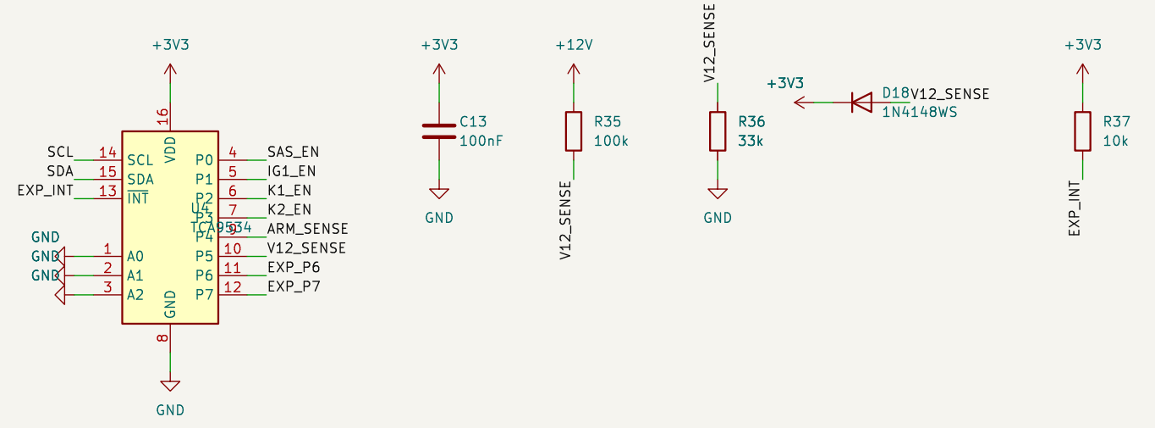
NEP (12 V pull–up) and VSP (5 V pull–up) open–drain outputs


D4_NEP
R24
100R
NEP_G
NEP_G
R25
100k
GND
NEP_DRV
NEP_G
Q8
2N7002
GND
+12V
R26
4.7k
NEP_DRV
NEP_DRV
R27
100R
NEP_STIM
D4_NEP
R28
1k
LED_NEP
GND
D14
LED_NEP
BLU
TX_VSP
R29
100R
VSP_G

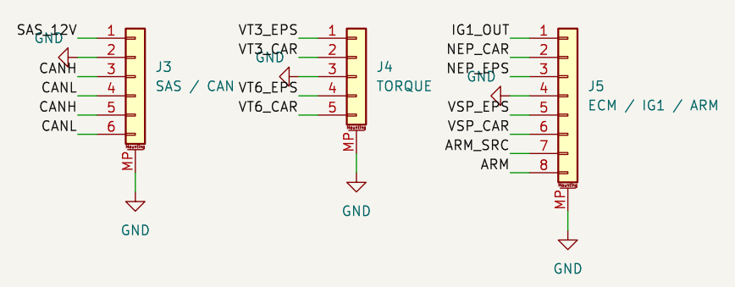
Torque DAC: MCP4728 (2.048 V ref, x1) –> MCP6004 x2.2 on 5 V –> 0–4.5 V


+3V3
SCL
SDA
DAC_LDAC
SCL
SDA
LDAC
RDY/BSY
VDD
VSS
U2
MCP4728
VOUTA
VOUTB
VOUTC
VOUTD
DAC_A
DAC_B
DAC_C
DAC_D
GND
DAC_LDAC
R34
10k
GND
+3V3
C10
100nF
GND
OPA_P
OPA_N
U3A
MCP6004
OPA_OUT
OPB_P
OPB_N
U3B

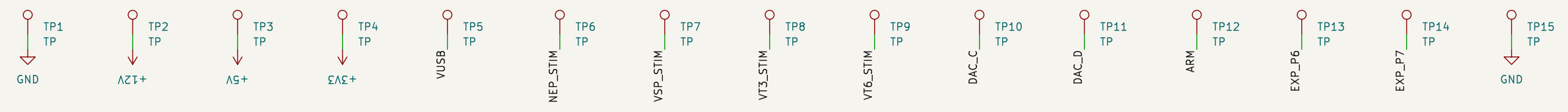
I2C expander TCA9534 (0x20): powers up as inputs


+3V3
SCL
SDA
EXP_INT
GND
GND
GND
SCL
SDA
INT
A0
A1
VDD
U4
TCA9534
P0
P1
P2
P3
P4
P5
P6
SAS_EN
IG1_EN
K1_EN
K2_EN
ARM_SENSE
V12_SENSE
EXP_P6
EXP_P7
+3V3
C13
100nF
GND
+12V
R35
100k
V12_SENSE
V12_SENSE
R36
33k
GND
+3V3
D18
V12_SENSE
1N4148WS
+3V3
R37
10k
EXP_INT

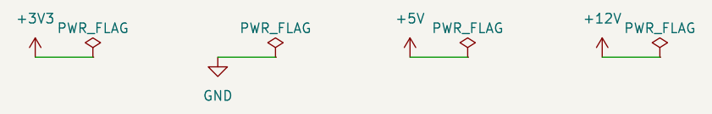
Connectors



Test pads


TP1
TP
GND
TP2
TP
+12V
TP3
TP
+5V
TP4
TP
+3V3
TP5
TP
VUSB
TP6
TP
NEP_STIM
TP7
TP
VSP_STIM
TP8
TP
VT3_STIM

Power: +3V3/GND from the Feather; +12V via D1; +5V from the buck (L1)


+3V3
PWR_FLAG
PWR_FLAG
GND
+5V
PWR_FLAG
+12V
PWR_FLAG

Generated by gen/make_sch.py from gen/design.py
Honda Insight EPS bench / in–car stimulus + 12 V power

Sheet: /
File: bench_stim_wing.kicad_sch

Title: Bench stimulus FeatherWing

Size: A0	Date: 2026–11–02	**Rev: 1.0**
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